

Project Design Phase

Module 1: Problem Solution Fit

Project Name: Import Data Using Transform Maps (Spreadsheet)

1. Problem Statement

In enterprise and organization-level data management, large volumes of records are routinely collected and updated across standalone spreadsheets (.xlsx or .csv). Manually copying and pasting or re-entering this spreadsheet data into destination database tables causes critical business challenges:

- **High Operational Overhead:** Manual data entry takes substantial time and human effort, slowing down administrative processes.
- **Human Errors & Inconsistencies:** Manual keying frequently causes typographical errors, incorrect field matching, and broken data formats.
- **Record Duplication:** Without automated unique-identifier validation, inserting spreadsheet rows manually results in duplicate records across destination tables.
- **Lack of Data Validation:** Loading unvalidated raw data directly into production systems risks corrupting target tables with broken schema values.

2. Target Audience & Stakeholders

- **System Administrators:** Responsible for performing routine bulk updates and maintaining data integrity across platform modules.
- **Database & Data Operations Engineers:** Tasked with migrating legacy datasets into centralized production systems.
- **Business Analysts & Departmental Leads:** Require accurate, updated database records without waiting on manual entry backlogs.

3. Problem-Solution Fit Matrix

Identified Pain Points	Proposed System Capability	Business Value & Outcome
Time-Consuming Manual Entry	Bulk Spreadsheet Ingestion	Automates data ingestion by uploading entire Excel/CSV files in a single step, saving time.
Format Mismatch & Field	Automated & Custom Field	Maps spreadsheet columns

Errors	Mapping	directly to target table schema fields, eliminating field alignment mistakes.
Duplicate Database Records	Coalesce Matching Engine	Automatically checks unique keys (e.g., User ID, Email) to update existing records instead of duplicating them.
Risk of Data Corruption	Intermediate Staging (Import Set Table)	Holds raw data in a temporary staging layer so it can be inspected and validated before writing to destination tables.

4. Summary of Fit

The **Import Data Using Transform Maps** solution directly addresses the operational friction of manual spreadsheet handling. By placing an intermediate staging layer and automated mapping engine between the spreadsheet and target database tables, the solution delivers accurate, deduplicated, and fully validated data migrations with minimal manual intervention.